Type MS · Using the cosine rule, including when it gives a quadratic

Answer 1 of 12 · Question 4 part whole · 5 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

4	$(2x+5)^2=(3x-1)^2+(5x)^2-2\times(3x-1)\times 5x\cos 60°$	M1A1
	$15x^2-21x-24(=0)\quad (5x^2-7x-8=0)$	A1
	$x=\dfrac{21\pm\sqrt{21^2+4\times15\times24}}{30}$	M1
	$x=2.1456.....\ (\text{or } -0.7456...)$	
	$\therefore x=2.15$	A1 **[5]**

M1	Use the cosine rule in either form. Rule to be correct either by quoting and using the general formula or by implication from a correct substitution.
A1	Correct substitution in their cosine rule.
A1	Simplify to obtain a 3TQ. Terms in any order. $=0$ may be missing
M1	Solve their 3TQ by formula (correct general formula or correct substitution for their equation) or completing the square. Reach a positive value for x. Negative need not be seen. **Calculator solutions:** Correct answer from correct equation scores M1A1, otherwise M0A0
A1cao	Correct value for x. **Must** be 3 sf Negative value (if shown) must be eliminated or positive clearly identified as the required value..

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 1 of 12 · Question 4 part whole · 5 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
	$36 \quad (\quad 36)$	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 2 of 12 · Question 3 part a · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
3(a)	$BC^2 = 10^2 + 8^2 - 2 \times 10 \times 8 \times \cos 100^\circ$ $BC = 13.848\ldots \approx 13.8$	M1 A1 [2]
(b)	(i) $\sin ABC = \frac{8\sin 100^\circ}{13.848\ldots} \Rightarrow ABC = 34.6752\ldots^\circ \approx 34.7^\circ$ (ii) $\angle ACB = 180^\circ - 100^\circ - 34.6752\ldots^\circ = 45.324\ldots^\circ \approx 45.3^\circ$	M1A1 B1FT [3]
	(i)(ii) $\cos ACB = \frac{8^2+13.848\ldots^2-10^2}{2\times 8\times 13.848\ldots} \Rightarrow ACB = 45.330\ldots$ $ABC = 180° - 100° - 45.330\ldots = 34.6699\ldots$	[M1A1 B1FT]
(c)	$\angle MBC = 34.675 \div 2 = 17.337\ldots^\circ$ and $\angle BMC = 180^\circ - 17.337^\circ = 117.33\ldots^\circ$ $MC = \frac{13.848\sin 17.337^\circ}{\sin 117.33^\circ} = 4.6293\ldots$ Area of BMC = $\frac{1}{2} \times 4.629 \times 13.848 \times \sin 45.324^\circ = 22.79\ldots \approx 22.8\ (\text{cm}^2)$	B1FT M1 M1A1 [4]
	ALT	
	$ABM = 34.675 \div 2 = 17.337\ldots^\circ$ and $AMB = 180 - 100 - 17.33\ldots = 62.66\ldots$ $BM = \frac{10\sin 100}{\sin 62.66\ldots} = 11.086\ldots$ Area of BMC = $\frac{1}{2} \times 13.848\ldots \times 11.086\ldots \times \sin 17.337\ldots = 22.87\ldots$ $\approx 22.9\ (\text{cm}^2)$	[B1FT M1 M1A1]
		Total 9 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 2 of 12 · Question 3 part a · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

ALT		
(b)(i)(ii)	M1	For applying cosine rule correctly to find angle ACB Accept any appropriate trigonometry.
	A1	For awrt 45.3° (angle ACB)
	B1FT	FT $180 - 100 - \textit{their}\ 45.330$ ... (angle ABC)
(c)	B1ft	For both angles MBC and BMC
	M1	For applying the sine rule using their angles and BC to find length MC Allow alternative correct methods to find MC.
	M1	For using Area $= \frac{1}{2}ab\sin C$ using their length and angles. $\textit{Their}\ MC \neq 4$
	A1	Awrt 22.8 or awrt 22.9
ALT		
(c)	B1ft	For both angles ABM and AMB
	M1	For applying sine rule using their angles to find length BM. Allow alternative correct methods to find BM.
	M1	For using Area $= \frac{1}{2}ab\sin C$ using their length and angles. $\textit{Their}\ MC \neq 4$ if used.
	A1	Awrt 22.8 or awrt 22.9

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 2 of 12 · Question 3 part a · 2 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 3 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

		$\left(-\frac{}{3n}\right) \Rightarrow \left(\frac{}{9n}\right)$ or $\left(-\frac{}{3n}\right) \Rightarrow -\left(\frac{}{9n^2}\right)$ then this is M0 Their equation must be either a quadratic or linear if they cancel an n This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3},\ A=\frac{1}{2}$ isw $n = 0$ seen. This is equivalent to cancelling n's
	A1	For the correct value of n **and** A
	ALT 1	**First two marks same as main method**
	M1	Substitutes n in terms of A into an equation in A $\frac{'-\frac{1}{3A}'\left('-\frac{1}{3A}'-1\right)(A)^2}{2}=\frac{5}{36}$ **Do not** allow a substitution of $n=-\frac{1}{3}$ for this mark
	dM1	For solving **their** equation in A to find a value for A Their equation must be either a quadratic or linear if they cancel an A This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3},\ A=\frac{1}{2}$
	A1	For the correct value of n **and** A
	ALT 2	**First two marks same as main method**
	M1	Substitutes nA into $\frac{(nA)^2-A(An)}{2!}=\frac{5}{36}$
	dM1	Solves their equation to find a value for A provided no errors introduced. This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3},\ A=\frac{1}{2}$
	A1	For the correct value of n **and** A

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 3 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

(b)	M1	Uses the **correct form** for the fourth term of a binomial expansion with their A and their n. You may see this in terms of A and n in part (a) Accept as a minimum, the correct power of $\left(\frac{1}{2}\right)^3$ with the correct denominator i.e. 3! Allow the presence of x^3 for this mark.
	A1	For the correct value of $-\frac{5}{81}$ If you see the correct value following the correct values of A and n with no working, award M1A1

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 3 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
3(a)	$36=\frac{1}{2}\times x\times(x+7)\times\sin 30^{\circ}\Rightarrow 36=\frac{x(x+7)}{4}\Rightarrow x^2+7x-144=0$	M1
	$\left[\text{Accept also e.g., } x^2+7x=144\right]$	
	$(x-9)(x+16)=0\Rightarrow x=9,(-16)\,(\text{cm})$ *	M1A1 cso [3]
(b)	$BC=\sqrt{9^2+16^2-2\times9\times16\cos 30^{\circ}}=9.3586...\approx 9.36\,(\text{cm})$	M1A1 [2]
(c)(i)	$\frac{\sin ABC}{16}=\frac{\sin 30^{\circ}}{9.3586}\Rightarrow \angle ABC=180^{\circ}-58.7408...\approx 121.3^{\circ}$	M1A1
(ii)	$\angle BCA=180^{\circ}-30^{\circ}-121.259^{\circ}=28.740...\approx 28.7^{\circ}$	B1ft [3]
		Total 8 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 3 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 5 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	States or uses $\theta = \frac{21-r}{r}$
	M1	Uses the correct formula for the area of a sector and substitutes their expression for θ
	A1 cso	For the correct expression for A exactly as written with no errors. That is $A = \frac{r}{2}(21-r)$ including $A =$.. However, if they have a string of working, then allow $A =$ at the top For example ., $A =$ ..... $=$.... $= \ldots\ldots\ldots\ldots = \frac{r}{2}(21-r)$ is acceptable
	ALT	
	B1	States $l = 21-r$
	M1	Uses the alternative formula $A = \frac{1}{2}rl$ and substitutes in their l
	A1 cso	For the correct expression for A exactly as written with no errors.
(b)	B1	Forms a correct inequality **and** a correct 3TQ (accept in any form) using the given expression for A and the value of 27
	M1	Solves their 3TQ by any method to obtain two values for r If they use a calculator to solve an incorrect 3TQ, award only if a method is seen. If the 3TQ is correct, and they use a calculator, award for correct roots seen.
	M1	Forms an inequality for the **inside** region using their values of r Allow any critical values which were obtained from their quadratic equation. Accept any acceptable form that defines a continuous **inside** region For example: $3 \leqslant r$ and $r \leqslant 18$ or $3 \leqslant r \cap r \leqslant 18$ Accept even $3 \leqslant r$ or $r \leqslant 18$
	A1	For the correct inequality only in this form. $3 \leqslant r \leqslant 18$
(c)	B1ft	For both $\theta = \frac{1}{6}$ and $\theta = 6$ You must follow through their values for r for this mark.
	B1	For the correct inequality only in this form.. $\frac{1}{6} \leqslant \theta \leqslant 6$

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 3 of 12 · Question 3 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Question	Scheme	Marks
5(a)	**Mark both parts together**	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 4 of 12 · Question 8 part b · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

8(a)	$AB^2 = 4^2 + 2^2,\ BC^2 = 2^2 + 6^2,\ AC^2 = 2^2 + 4^2$	M1 (any one)
	(i) $AB = \sqrt{20}$ (4.47) (ii) $BC = \sqrt{40}$ (6.32) (iii) $AC = \sqrt{20}$ (4.47) or equivalents	A1A1A1 (4)
(b)	Any complete method for finding one of the angles: eg $AB^2 + AC^2 = BC^2 \Rightarrow \angle A = 90°$ or use trigonometry	M1
	$\angle A = 90°,\ \angle B = \angle C = 45°$	A1, A1 (3)
I	(centre at midpoint of BC) $(5,5)$	M1A1 (2)
(d)	Radius $= \frac{1}{2}BC = \frac{1}{2}\sqrt{40} = \sqrt{10}$ (Working for (d) may be seen in a previous part)	M1A1 (2) [11]

(a)	
M1	Use Pythagoras with a plus sign to obtain AC^2, BC^2 or AC^2. If the answer is incorrect it must be clear that the correct coordinates have been used correctly.
A1A1A1	Award A1 for each correct length. Ignore labels (i), (ii) and (iii). Award M1A1A1A1 / M1A1A1A0 / M1A1A0A0 as appropriate.
SC:	If there is no working shown but at least one length is correct, award M1 and deduct one A mark for each incorrect length. (no length correct and no working ⇒ M0) If all 3 lengths are correct to at least 3 sf, award M1A1A1A0 If 2 are correct to at least 3 sf, award M1A1A0A0
(b)	
M1	Attempt to obtain any of the required angles. Method must be complete (ie reach a value for one angle) and formula used must be correct and values must be substituted into a correct formula.
A1	$\angle A = 90°$ Any labelling given can be ignored.
A1	$\angle B = \angle C = 45°$ All 3 correct w/o working scores M1A1A1
(c)	
M1	For indicating that the centre is at the midpoint of BC. This can be stated explicitly or used by attempting to find the midpoint. **OR:** Find equations for perpendicular bisectors of 2 of the sides and find the point of intersection
A1	Both coordinates correct. Correct answer written down w/o working scores M1A1
(d)	
M1	For indicating that the radius is half the length of BC. This can be stated explicitly or used by attempting to find half of their BC (not nec in the required form).
A1	Correct length of the radius, in the required form.
NB	If half the length of BC has been found earlier the marks for (d) can only be awarded if the length of the radius has been written in (d).

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 4 of 12 · Question 8 part b · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{1}{1}$	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 5 of 12 · Question 3 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

3 (a)	$\cos(\angle B) = \dfrac{10^2+8^2-12^2}{2\times10\times8} = \dfrac{10^2+6^2-x^2}{2\times10\times6}$	M1 M1
	$\dfrac{20}{160} = \dfrac{136-x^2}{120}$	
	$x^2 = 121$	dM1
	$x = 11$ *	A1 cso (4)
	Alternative Method	
	$\cos(ADB) = \dfrac{x^2+6^2-10^2}{2\times6\times x} = \dfrac{x^2-64}{12x}$	M1
	$\cos(ADC) = \dfrac{x^2+2^2-12^2}{2\times2\times x} = \dfrac{x^2-140}{4x}$	M1
	$\dfrac{x^2-64}{12x} + \dfrac{x^2-140}{4x} = 0$	
	$4x^2 = 484$	dM1
	$x = 11$ *	A1 cso (4)
(b)	Angle ABC / $ABD = \cos^{-1}\left(\dfrac{10^2+8^2-12^2}{2\times10\times8}\right) = 82.8^\circ$	B1 M1
	$\dfrac{1}{2}\times10\times6\times\sin 82.8 = 29.8$	M1 A1 (4) **[8]**
	Alternative Method 1	
	Angle $ADB = \cos^{-1}\left(\dfrac{11^2-64}{12\times11}\right) = 64.42^\circ$	B1 M1
	$\dfrac{1}{2}\times6\times11\times\sin 64.42 = 29.8$	M1 A1 (4)
	Alternative Method 2	
	$\cos ABC = \dfrac{10^2+8^2-12^2}{2\times10\times8} = \dfrac{1}{8}$	B1
	$\sin ABC$ / $ABD = \sqrt{1-\left(\dfrac{1}{8}\right)^2} = \sqrt{\dfrac{63}{64}} = \dfrac{3}{8}\sqrt{7}$	M1
	$\dfrac{1}{2}\times10\times6\times\dfrac{3}{8}\sqrt{7} = 29.8$	M1 A1 (4)

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 5 of 12 · Question 3 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	Notes
(a)	
M1	Use of cosine rule to obtain a correct expression for $\cos(\angle B)$. The correct formula in either form may be used.
M1	Use of cosine rule to obtain a second correct expression for $\cos(\angle B)$. The correct formula in either form may be used.
dM1	Dependant on previous M mark - for solving leading to $x^2 = ...$
A1 cso	For obtaining the given result
	Alternative
M1	Use of cosine rule to obtain a correct expression for $\cos(ADB)$. The correct formula in either form may be used.
M1	Use of cosine rule to obtain a correct expression for $\cos(ADC)$. The correct formula in either form may be used.
dM1	Dependant on previous M mark - for solving leading to $4x^2 = ...$
A1	For obtaining the given result
(b)	
B1	For use of the cosine rule to find angle B
M1	For $\cos^{-1}\left(\frac{10^2+8^2-12^2}{2\times10\times8}\right)$ oe
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8
	Alternative 1
B1	For use of the cosine rule to find angle ADB
M1	For $\cos^{-1}\left(\frac{11^2-64}{12\times11}\right)$ oe
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8
	Alternative 2
B1	For use of the cosine rule to find $\cos B$
M1	Use of $\sin^2 A + \cos^2 A = 1$
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 5 of 12 · Question 3 part a · 4 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 6 of 12 · Question 9 part a · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1	(2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1	
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1	
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$		
	$AC = 6\sqrt{6}$	A1	(4)
(c)	$\dfrac{\sin(\theta^\circ + \phi^\circ)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta^\circ + \phi^\circ)}{x} = \dfrac{\sin BCD}{6}$	M1A1	
	$\dfrac{\sin\phi^\circ}{x} = \dfrac{\sin BCD}{6}$	M1	
	$\therefore \sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$	A1	(4)
(d)	$\sin(\theta^\circ + \phi^\circ) = \sin\phi^\circ \Rightarrow (\theta + \phi) = 180 - \phi$ $(\text{or } \phi \text{ or } 360 + \phi \text{ (not possible)})$	M1	
	$\therefore \theta = 180 - 2\phi$	A1	(2)
			[12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for cos *BAD* and cos *BAC*: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 6 of 12 · Question 9 part a · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2 + x^2 - 6^2}{2\times 6\times x}$ and $\cos BCA = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x}$
	M1	$\frac{6^2 + x^2 - 6^2}{2\times 6\times x} = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x} \Rightarrow x^2 = 2x^2 - 54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ + \phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ + \phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ + \phi^\circ) = \frac{x \sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta + \phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta + \phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ) + \sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta + \phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta + \phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ + \phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ + \phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ + \phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180 - \phi)^\circ \Rightarrow \theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$
	A1	For rearranging $\theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$ to achieve $\therefore \theta = 180 - 2\phi$ This is a show question and there must be no errors here.

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 6 of 12 · Question 9 part a · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 7 of 12 · Question 7 part a · 2 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

7 (a)	**Throughout this question condone missing degree signs** $\cos\theta^\circ = \frac{6^2+8^2-k^2}{2\times6\times8} = \frac{100-k^2}{96}$ *	M1A1cso [2]
(b)	$\sqrt{455} = \frac{1}{2}\times6\times8\times\sin\theta^\circ$ $\Rightarrow \sin\theta^\circ = \frac{\sqrt{455}}{24} \Rightarrow \left(\sin^2\theta^\circ = \frac{455}{576}\right)$	M1 A1
	$\cos^2\theta^\circ = 1-\sin^2\theta^\circ \Rightarrow \cos^2\theta^\circ = 1-\frac{455}{576} = \frac{121}{576}$ $\Rightarrow \cos\theta^\circ = \pm\frac{11}{24}$ both values required $\cos\theta^\circ = \frac{11}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56} = \left(2\sqrt{14}\right)$ $\cos\theta^\circ = -\frac{11}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 144 \Rightarrow k = 12$	M1 A1 M1A1 A1
	ALT $\sqrt{455} = \frac{1}{2}\times6\times8\times\sin\theta^\circ$ $\Rightarrow \sin\theta^\circ = \frac{\sqrt{455}}{24}$ $\theta^\circ = \sin^{-1}\left(\frac{\sqrt{455}}{24}\right)$ $(= 62.72\ldots^\circ)$ $\theta^\circ = 62.72\ldots, 117.27\ldots$ both values required $\cos\theta^\circ = \frac{100-k^2}{96} \Rightarrow k^2 = 100-96\cos\theta^\circ$ $\Rightarrow k^2 = 100-96\cos 62.72\ldots \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta^\circ = \frac{100-k^2}{96} \Rightarrow k^2 = 100-96\cos\theta^\circ$ $\Rightarrow k^2 = 100-96\cos 117.27\ldots \Rightarrow k^2 = 144 \Rightarrow k = 12$	[7] [M1 A1 M1 A1 M1A1 A1]
		Total 9 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 7 of 12 · Question 7 part a · 2 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

(b)	**M1**	For using the **correct** formula for area of a triangle and substitution of the given values to obtain $\sqrt{455} = \frac{1}{2} \times 6 \times 8 \times \sin\theta°$ and attempt to rearrange to obtain $\sin\theta° = \cdots$
	A1	$\sin\theta° = \frac{\sqrt{455}}{24}$ Allow for $\sin\theta = \frac{\sqrt{455}}{0.5\times8\times6}$
	M1	For use of $\sin^2\theta + \cos^2\theta = 1$ to obtain a value for $\cos\theta°$. $\cos^2\theta° = 1 - \frac{'455'}{576} \Rightarrow \cos\theta° = \pm\sqrt{1 - \frac{'455'}{576}}$ Allow use of their $\sin\theta°$ provided $-1 \leq \sin\theta° \leq 1$ Allow if only one value of $\cos\theta°$ obtained.
	A1	$\cos\theta° = \pm\frac{11}{24}$
	M1	For forming an equation for k using their $\cos\theta°$ and attempt to solve for k. $\cos\theta° = \frac{'11'}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta° = '-\frac{11'}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 144 \Rightarrow k = 12$
	A1	For one correct value of k $\sqrt{56}$ or awrt 7.48 **or** awrt 12
	A1	For both correct values of k $\sqrt{56}$ or awrt 7.48 **and** awrt 12
ALT – working with angles		
	M1	For using the **correct** formula for area of a triangle and substitution of the given values to obtain $\sqrt{455} = \frac{1}{2} \times 6 \times 8 \times \sin\theta°$ and attempt to rearrange to obtain $\sin\theta° = \cdots$
	A1	$\sin\theta° = \frac{\sqrt{455}}{24}$ Allow for $\sin\theta = \frac{\sqrt{455}}{0.5\times8\times6}$
	M1	For use of the inverse trigonometric function to obtain a value for $\theta°$ $\theta° = 62.72\ldots, 117.27\ldots$ If working not shown then award for angle correct to a minimum of 1 d.p. Allow if only one value of $\theta°$ found. Condone working in radians awrt 1.09, awrt 2.05
	A1	$\theta° = 62.72\ldots, 117.27\ldots$ Allow awrt 62.7°, 117.3° Both angles found. Condone working in radians awrt 1.09, awrt 2.05
	M1	For forming an equation in k using their θ and an attempt to solve for k. $\cos\theta° = \frac{100-k^2}{96} \Rightarrow k^2 = 100 - 96\cos\theta°$ $\Rightarrow k^2 = 100 - 96\cos 62.72\ldots \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta° = \frac{100-k^2}{96} \Rightarrow k^2 = 100 - 96\cos\theta°$ $\Rightarrow k^2 = 100 - 96\cos 117.27\ldots \Rightarrow k^2 = 144 \Rightarrow k = 12$
	A1	For one correct value of k $\sqrt{56}$ or awrt 7.48 **or** awrt 12
	A1	For both correct values of k and no others. $\sqrt{56}$ or awrt 7.48 **and** awrt 12

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 7 of 12 · Question 7 part a · 2 marks

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Question number	Scheme	Marks

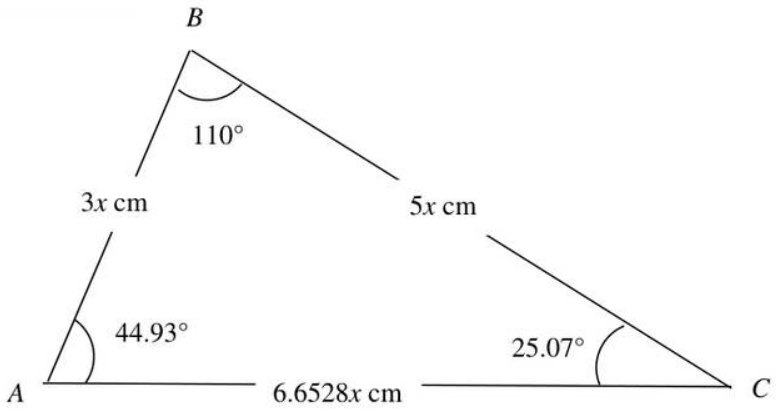
Type MS · Using the cosine rule, including when it gives a quadratic

Answer 8 of 12 · Question 2 part a · 4 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

2(a)	$AC = \sqrt{(3x)^2 + (5x)^2 - 2\times 3x\times 5x\times\cos 110^\circ} = 6.6528...x$	M1A1
	$\sin\angle BCA = \dfrac{3x\sin 110^\circ}{'6.6528'x} = 0.4237... \Rightarrow \angle BCA = 25.07...^\circ$	M1
	Accept awrt 25.1°	A1 [4]
	ALT (for 2nd M1 A1)	
	$\cos\angle BCA = \dfrac{(5x)^2 + ('6.6528'x)^2 - (3x)^2}{2\times 5x\times '6.6528'x} = 0.9057... \Rightarrow \angle BCA = 25.07^\circ$	[M1
	Accept awrt 25.1°	A1]
(b)	$24 = \dfrac{1}{2}\times 3x\times 5x\times\sin 110^\circ$	M1
	$\Rightarrow x = \sqrt{\dfrac{24\times 2}{3\times 5\times\sin 110^\circ}} = (1.84536...)$	M1
	Accept awrt 1.85	A1 [3]
		Total 7 marks

Useful sketch



Type MS · Using the cosine rule, including when it gives a quadratic

Answer 8 of 12 · Question 2 part a · 4 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using a correct cosine rule for AC or AC^2. Note, this is given in the formula sheet so must be correct for this mark.
	A1	For finding a length of $AC = 6.6528...x$ or $AC = \sqrt{44.26....x^2}$ accept awrt $6.7x$ or $\sqrt{44.3x^2}$ **Note:** Allow the missing x here and throughout their working This mark can be implied by sight of $AC = \sqrt{34x^2 - 30x^2\cos 110}$ if they carry it through to find the angle in the next step. For example: $\sin\angle BCA = \dfrac{3x\sin 110^\circ}{\sqrt{34x^2 - 30x^2\cos 110}}$
	M1	For using any appropriate trigonometry to find the size of angle BCA. For example; **Sine Rule** $\sin\angle BCA = \dfrac{3x\sin 110^\circ}{'6.6528'x} = 0.4237... \Rightarrow \angle BCA = 25.07...^\circ$ **Cosine Rule** $\cos\angle BCA = \dfrac{(5x)^2 + (6.6529x)^2 - (3x)^2}{2\times 5x\times 6.6529x} \Rightarrow \angle BCA = 25.07$ Allow a missing x from their working provided it is consistent. Do not allow for example $\sin\angle BCA = \dfrac{3\boldsymbol{x}\sin 110^\circ}{'6.6528'}$
	A1	For awrt 25.1°
	ALT – Uses sine rule	
	M1	Use of sine rule: $\dfrac{\sin(70-\theta)}{5x} = \dfrac{\sin\theta}{3x} \Rightarrow \left[\dfrac{\sin(70-\theta)}{\sin\theta} = \dfrac{5}{3}\right]$
	A1	Expands $\sin(70-\theta) = \sin 70\cos\theta - \cos 70\sin\theta$
	M1	Uses the tan identity and rearranges to make $\tan\theta$ the subject. $\dfrac{\sin 70\cos\theta - \cos 70\sin\theta}{\sin\theta} = \dfrac{5}{3}$ $\Rightarrow \dfrac{\sin 70}{\tan\theta} - \cos 70 = \dfrac{5}{3} \Rightarrow \tan\theta = \dfrac{\sin 70}{\cos 70 + \dfrac{5}{3}}$
	A1	For awrt 25.1°
(b)	M1	For using the correct formula for the area of a triangle with the correct angle and side lengths. e.g., Can use AB and AC with $\angle ABC = 110^\circ$ or BC and AC with their $\angle BCA$ [25.1°] or AB and AC with their $(180^\circ - 110^\circ - 25.07^\circ = 44.93^\circ)$

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 8 of 12 · Question 2 part a · 4 marks

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	M1	For finding a value of x using their value for AC and BCA or BAC The processing must be correct for this mark and they must be finding the square root of x for this mark.
	A1	For awrt 1.85

Question	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 9 of 12 · Question 2 part whole · 4 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Number		
2	$(2x-1)^2=(2x+4)^2+(x+2)^2-2(2x+4)(x+2)\cos 60^\circ$	M1
	$=x^2-16x-11=0$	M1
	$x=\dfrac{16\pm\sqrt{16^2+4\times 11}}{2}=8\pm 5\sqrt{3}$	M1
	[If $x=8-5\sqrt{3}$ then $2x-1$ is negative, so] $x=8+5\sqrt{3}$	A1 (4)
		Total 4 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 9 of 12 · Question 2 part whole · 4 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks
	$a^2 = \left(\ldots\right)^2 + 10^2 - \ldots$	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 10 of 12 · Question 5 part a · 5 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

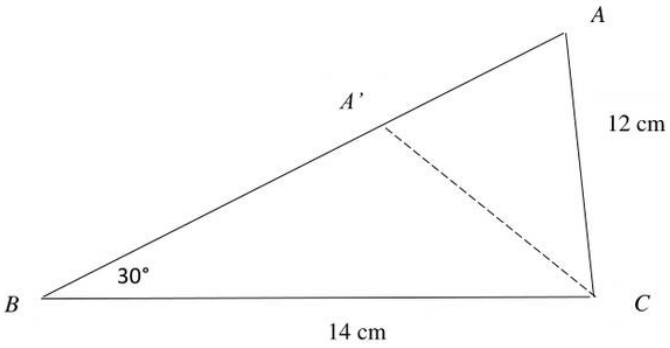
5(a)	$12^2 = 14^2 + x^2 - 2\times 14x\cos 30$	M1
	$x^2 - 14\sqrt{3}x + 52 = 0$	M1 A1
	$x = \dfrac{14\sqrt{3} \pm \sqrt{(14\sqrt{3})^2 - 4(1)(52)}}{2(1)}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$	A1cso [5]
(b)	Area $= \dfrac{1}{2}\times 14\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \sin 30$	M1
	or uses Area $= \dfrac{1}{2}\times 12\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \dfrac{7}{12}$	
	Difference in areas	
	$D = \dfrac{1}{2}\times 14\times\left(7\sqrt{3} + \sqrt{95}\right)\times \sin 30 - \dfrac{1}{2}\times 14\times\left(7\sqrt{3} - \sqrt{95}\right)\times \sin 30$	dM1
	$D = 7\sqrt{95}$	A1 [3]
ALT Part (a) – sine rule		
	$\dfrac{12}{\sin 30} = \dfrac{14}{\sin A} \Rightarrow \sin A = \dfrac{7}{12}$	{M1
	$\left(\dfrac{7}{12}\right)^2 + \cos^2 A \equiv 1 \Rightarrow \cos A = \sqrt{1 - \left(\dfrac{7}{12}\right)^2}$	M1
	$\sin A = \dfrac{7}{12}, \cos A = \pm\dfrac{\sqrt{95}}{12}$	A1
	$x = 12\cos A + 14\cos 30 \Rightarrow x = 12\times\dfrac{\pm\sqrt{95}}{12} + 14\times\dfrac{\sqrt{3}}{2}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$ *	A1 Cso [5]}
		Total 8 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 10 of 12 · Question 5 part a · 5 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

USEFUL SKETCH



Part	Mark	Additional Guidance
(a)	M1	Use the cosine rule to form an equation. Rule to be correct but may be in any rearranged form.
	M1	Rearrange to obtain a 3TQ. Terms in any order. = 0 may be missing.
	A1	Correct 3TQ Allow poor notation – missing = 0
	M1	Attempt to solve their 3TQ by use of the formula or completing the square. See general guidance for what constitutes an attempt to solve. Use of calculators: If the 3TQ is correct, accept correct solutions without working.
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$
(b)	M1	For use of area of triangle $= \frac{1}{2}ac\sin B$ with $a = 14$, $B = 30°$ and *their* x May use either value found for x. Or uses $= \frac{1}{2}bc\sin A$ with $c = 12$ and $\sin A = \frac{7}{12}$
	dM1	For finding the difference between their two areas provided both areas are positive This M mark is dependent on the previous M mark
	A1	Correct difference in areas (must be exact value)

	ALT Part (a) – Uses sine rule	
	M1	Use of the sine rule to find sin$A$
	M1	Use of $\sin^2 A + \cos^2 A \equiv 1$ to find an exact value for $\cos A$
	A1	Correct exact values for sin$A$ and $\cos A$
	M1	Method to find x $x = 12\cos A + 14\cos 30$ Allow *their* $\cos A$
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 10 of 12 · Question 5 part a · 5 marks

2025 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 11 of 12 · Question 3 part a · 3 marks

2025 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks
3 (a)	$15^2 = 12^2 + x^2 - 2\times 12\times x\times \cos 60^\circ$	M1
	$\Rightarrow 225 = 144 + x^2 - 12x \Rightarrow x^2 - 12x - 81 = 0$ *	M1A1 cso [3]
	Special Case in this part for using Sine Rule **The working must be CORRECT for the award of these marks.**	
	$\frac{15}{\sin 60} = \frac{12}{\sin C} \Rightarrow C = 43.9^\circ$ **AND** $\frac{15}{\sin 60} = \frac{x}{\sin 76.1..}$ $\Rightarrow x = 16.8(16...), -4.8(16...)$ **Both roots required**	B1
	$x^2 - 12x - 81 = 0 \Rightarrow x = 16.8...,\ \ 4.81...$ and concludes that x must satisfy the given quadratic equation	B1
(b)	$x^2 - 12x - 81 = 0 \Rightarrow x = 6 + \sqrt{117} = \left[6 + 3\sqrt{13}\right]$	M1
	$A = \frac{1}{2}\times 12\times \left(6 + 3\sqrt{13}\right)\times \sin 60^\circ$	M1
	$A = 3\sqrt{3}\left(6 + 3\sqrt{13}\right) = 9\left(2\sqrt{3} + \sqrt{39}\right)$	A1 [3]
		Total 6 marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 11 of 12 · Question 3 part a · 3 marks

2025 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 12 of 12 · Question 9 part b · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

9(a)	$\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$	M1A1	(2)
(b)	$\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$	B1	
	$ADB = 180 - BDC \Rightarrow -\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$	M1A1	
	$2x^2 = 108 \Rightarrow x = 3\sqrt{6}$		
	$AC = 6\sqrt{6}$	A1	(4)
(c)	$\dfrac{\sin(\theta° + \phi°)}{2x} = \dfrac{\sin BCD}{12} \Rightarrow \dfrac{\sin(\theta° + \phi°)}{x} = \dfrac{\sin BCD}{6}$	M1A1	
	$\dfrac{\sin \phi°}{x} = \dfrac{\sin BCD}{6}$	M1	
	$\therefore \sin \phi° = \sin(\theta° + \phi°)$	A1	(4)
(d)	$\sin(\theta° + \phi°) = \sin \phi° \Rightarrow (\theta + \phi) = 180 - \phi$ $(\text{or } \phi \text{ or } 360 + \phi \text{ (not possible)})$	M1	
	$\therefore \theta = 180 - 2\phi$	A1	(2)
			[12]

Part	Mark	Notes
(a)	M1	For using a correct cosine rule $\cos ADB = \dfrac{6^2 + x^2 - 12^2}{2\times 6x} = \dfrac{x^2 - 108}{12x}$ or $12^2 = x^2 + 6^2 - 2\times 6\times x\cos ADB$
	A1	Simplifies to $\cos ADB = \dfrac{x^2 - 108}{12x}$
(b)	B1	For the correct expression $\cos BDC = \dfrac{6^2 + x^2 - 6^2}{2\times 6x} = \dfrac{x^2}{12x}$
	M1	$\cos BDC = -\cos ADB$ and $\cos BDC = -\dfrac{x^2}{12x}$ so $-\dfrac{x^2}{12x} = \dfrac{x^2 - 108}{12x}$ and attempts to solve
	ALT 1 – uses triangles *BAD* and *BAC*	
	B1	For **both** of the following correct expressions for cos *BAD* and cos *BAC*: $\cos BAD = \dfrac{12^2 + x^2 - 6^2}{2\times 12\times x}$ and $\cos BAC = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x}$
	M1	$\angle BAC = \angle BAD$ so equates their two expressions $\dfrac{12^2 + x^2 - 6^2}{2\times 12\times x} = \dfrac{12^2 + (2x)^2 - 6^2}{2\times 12\times 2x} \Rightarrow \dfrac{108 + x^2}{24x} = \dfrac{108 + 4x^2}{48x}$ and attempts to solve
	ALT 2 – uses triangles *BCD* and *BCA*	

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 12 of 12 · Question 9 part b · 4 marks

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	B1	For **both** of the following correct expressions for cos BAD and cos BAC: $\cos BCD = \frac{6^2 + x^2 - 6^2}{2\times 6\times x}$ and $\cos BCA = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x}$
	M1	$\frac{6^2 + x^2 - 6^2}{2\times 6\times x} = \frac{6^2 + (2x)^2 - 12^2}{2\times 6\times 2x} \Rightarrow x^2 = 2x^2 - 54$ and attempts to solve
	Final A marks for all three methods	
	A1	For the correct value of $x = 3\sqrt{6}$
	A1	For $AC = 6\sqrt{6}$
(c)	M1	Uses sine rule on triangle ABC: $\frac{\sin(\theta^\circ + \phi^\circ)}{2x} = \frac{\sin BCD}{12} \Rightarrow \frac{\sin(\theta^\circ + \phi^\circ)}{x} = \frac{\sin BCD}{6}$
	A1	Achieves the correct expression for $\sin(\theta^\circ + \phi^\circ) = \frac{x \sin BCD}{6}$
	M1	Uses sine rule on triangle BDC: $\frac{\sin\phi^\circ}{x} = \frac{\sin BCD}{6} \Rightarrow \left(\sin\phi^\circ = \frac{x\sin BCD}{6}\right)$
	A1	Shows that $\sin\phi^\circ = \sin(\theta^\circ + \phi^\circ)$ with no errors
	ALT 1 – Uses exact values for the trigonometric ratios and the expansion for sin $(A + B)$	
	M1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **or** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **or** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	A1	Finds $\cos\theta = \frac{7}{8} \Rightarrow \sin\theta = \frac{\sqrt{15}}{8}$ **and** $\cos\phi = \frac{1}{4} \Rightarrow \sin\phi = \frac{\sqrt{15}}{4}$ Accept $\theta = 28.95...^\circ$ **and** $\phi = 75.52...^\circ \Rightarrow \sin\phi = 0.968...$
	M1	Expands $\sin(\theta + \phi) = \frac{\sqrt{15}}{8}\times\frac{1}{4} + \frac{\sqrt{15}}{4}\times\frac{7}{8} = \left(\frac{\sqrt{15}}{4}\right)$ Or $\sin(\theta + \phi)^\circ = \sin(28.95^\circ)\cos(75.52^\circ) + \sin(75.52^\circ)\cos(28.95^\circ) = 0.968... = \sin\phi^\circ$
	A1	Shows that $\sin(\theta + \phi) = \frac{\sqrt{15}}{4}$ and $\sin\phi = \frac{\sqrt{15}}{4}$ so $\sin(\theta + \phi) = \sin\phi$ with no errors. If they use approximate values for sin θ and ϕ withhold this final mark so A0
	ALT 2 – Uses $\angle BCD = 52.2...^\circ$	
	M1	Finds $\angle BCD = 52.2...^\circ$ using cosine rule and applies sine rule on triangle ABC $\frac{\sin(\theta^\circ + \phi^\circ)}{6\sqrt{6}} = \frac{\sin 52.2^\circ}{12}$
	A1	Shows that $\sin(\theta^\circ + \phi^\circ) = 0.968...$
	M1	Uses sine rule on triangle BD: $\frac{\sin\phi^\circ}{3\sqrt{6}} = \frac{\sin 52.2^\circ}{6} \Rightarrow \sin\phi^\circ = 0.968... = \sin(\theta^\circ + \phi^\circ)$
	A1	If they use an approximate value for angle BCD withhold this final mark so A0
(d)	M1	For writing $\sin\phi^\circ = \sin(180 - \phi)^\circ \Rightarrow \theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$
	A1	For rearranging $\theta^\circ + \phi^\circ = 180^\circ - \phi^\circ$ to achieve $\therefore \theta = 180 - 2\phi$ This is a show question and there must be no errors here.

Type MS · Using the cosine rule, including when it gives a quadratic

Answer 12 of 12 · Question 9 part b · 4 marks

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Question Number	Scheme	Marks